Roshan Raj

Curriculum Vitae

Upto Feb. 2024

Personal Details

Name:	Roshan Raj
Nationality:	Indian
Present address:	SVNIT, Surat, Gujarat, India-395007
Webpage:	kaalkrit.github.io
Email ID:	roshankaalkrit@gmail.com, i19ph002@phy.svnit.ac.in

Research Interests

I wanted to research the mathematical aspects of Quantum Field Theories and Foundations of Quantum Theory. Fundamentally passionate about getting the essence of Time, Mass, Charge, and Symmetries in the Universe and aspiring to engage in Unification Theories and interpretative endeavours over time. I am also interested in exploring string theory and holographic dualities.

Educational Details

2019-24	Five Years Integrated M.Sc. (Physics), *Sardar Vallabhbhai* National Institute of Technology (SV-NIT), Surat, India. CGPA: 9.33/10 (current) Title of Dissertation: *Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum**. Supervisor: Dr. Vikash K. Ojha.
2016-18	Senior Secondary High Schooling, Loyola High School, CBSE, Patna, Bihar, 77.2% [Physics, Chemistry, Mathematics, Informatics Practices, English]
2010-16	High Schooling, St. Paul's High School, ICSE, Patna, Bihar, 91.6%, [Science, Mathematics, Computer App., Social Sci., Hindi, English]

Research Experiences

[11]
Tentative Date of completion: May 2024
Thesis Title: `Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum`
Guide: Dr Vikash K Ojha, SVNIT-Surat
Type: Master's Dissertation (Final)
Abstract Draft: To compile the Part I and Part II work on relativistic quantum mechanics. To extend the work by finding the eigenvalues of the relativistic anharmonic oscillator by algebraic methods (creation-annihilation operators). To apply the alternate kinetic energy expression (relativistic) and the Kinetic Energy and Momentum relationship in classical theory to determine the relativistic expression of density of states, etc. Finally, all non-relativistic results (irrespective of classical or quantum theory) can be deduced by substituting the Lorentz factor $\gamma = 1$.

[10]
Tentative Date of completion: Last week of Feb 2024
Title: `On the novel approach to Relativistic Quantum Mechanics-II`
Guide: Dr Vikash K Ojha, SVNIT-Surat
Type: Master's Dissertation-Mid-Semester (Part II/B)
Abstract: Einstein's equations define the link between total energy and momentum for fast-moving particles,

but not a simple relationship for their kinetic energy. A more refined relativistic approach allows the formulation of a special quantum wave equation. Solving this equation for a simple system (harmonic oscillator) reveals interesting behaviours: energy levels get squished, the probability of finding the particle increases, and the size of a hydrogen atom shrinks as the electron speeds up. The research delves into further implications, including a revised description of the electron's magnetic moment, ensuring the equation works under different reference frames (Lorentz covariance) and confirms that particle number remains constant and recovers standard results for slow particles.

[9]
Date of completion: 04 Dec 2023
Title: `On the novel approach to Relativistic Quantum Mechanics-I`
Guide: Dr Vikash K Ojha, SVNIT-Surat
Type: Master's Dissertation-Preliminary (part I/A)
URL: **Click to See**
Abstract: A connection between the kinetic energy and momentum for a relativistic system is derived along with an alternate expression of relativistic kinetic energy and finally used in quantum mechanical problems to demonstrate that it can formulate a consistent relativistic quantum mechanical Hamiltonian. The work seems essential as it shows a compressible interpretation of probability density. It is noticed that the applications of the *relativistic Schrodinger's equation result in predicting compression of Energy levels with increasing velocity* and increased probability density for the increasing value of velocity in ideal systems such as one-dimensional quantum mechanical problems. *The work seems vital as it resolves Klein's paradox.* Finally, one can obtain non-relativistic results by substituting the Lorentz factor $\gamma = 1$ for all required cases.

[8]
Date of completion: 05 Jul 2023
Date of pre-print/publication: 30 Oct 2023: OSF Preprints
Title: `Obtaining the Klein-Gordon wave equation without using the quantum operators`
Type: Article
DOI: **10.31219/osf.io/6hmyf**
Abstract: The derivation of a wave equation (Klein-Gordon) for relativistic particles is presented, highlighting the difference between it and the governing law. This equation shares similarities with the Maxwell wave equation, and for massless particles, the wave velocity matches light, suggesting the wave-particle duality of light.

[7]
Date of completion: 19 Nov 2022
Title: `New light on the concepts of Observable and indeterminacy in the quantum realm`
Guide: Dr Vikash K Ojha, SVNIT-Surat
Type: UG Project III, [Jul-Dec, 2022]
URL: **Click to See**
Abstract: The work deals with the two objectives of studying (a) providing insights into observable-unobservable correspondence and (b) presenting that there exists a complement form of the Born-Jordan quantum condition. Complement form can be derived if the mathematical structure of an observable is considered slightly different from Heisenberg's quantum theoretical quantities. While obtaining the complement form, it was realised that the imaginary factor (say, $i = \sqrt{-1}$) seemingly acted as an operator and appeared to transform conventionally accepted observables into non-observables and vice versa. One of the significant findings introduces a similar uncertainty principle that exists with the Heisenberg uncertainty principle.

[6]
Date of completion: 19 Nov 2022
Date of pre-print/publication: 08 Oct 2023: OSF Preprints
Title: `An open-ended story on quantization`
Type: Article
DOI: **10.31219/osf.io/92bvp**
Abstract: To provide a write-up, impressing on the evolutionary track of quantization of physical quantities such as Energy, momentum, and angular momentum. In addition, to question which fundamental quantity is responsible for making these derived quantities quantized.

[5]
Date of completion: 17 Sep 2022
Date of pre-print/publication: 10 Oct 2023: OSF Preprints
Title: `Reviewing Observables in Classical and Quantum Mechanics`
Guide: Dr Vikash K Ojha, SVNIT-Surat

Type: Article
DOI: **10.31219/osf.io/p2ufx**
Abstract: This explores how measuring things in the quantum world (observables) differs from classical measurements. Heisenberg (1925) showed some things are inherently unobservable in the quantum realm, but their "essence" can be grasped through expectation values. By reinterpreting analogies, classical-like information might be retrieved and Interestingly, connections might exist between the two through reinterpretation.

[4]
Date of completion: 17 Jul 2022
Title: `The pictures of quantum dynamics` [*Pramatrik Gatiki ke Chitra*]
Guide: Dr Vikash K Ojha, SVNIT-Surat
Type: Internship report [May-Jul, 2022]
URL: **Request Book Chapter**
Abstract: This Hindi chapter explores 3 quantum dynamics methods, drawing on classical physics, quantum mechanics, and philosophy. Intended for a general audience, it aims to educate the public about quantum dynamics and its historical context.

[3]
Date of completion: 09 May 2022
Date of pre-print/publication: 21 Sep 2022: arXiv Preprints
Title: `On the investigation of two non-neutral static bodies`
Guide: Dr Sharad K Yadav, SVNIT-Surat
Type: UG Project II, [Jan-May, 2022]
DOI: **10.48550/arXiv.2209.10641**
Abstract: This work studies the statics of two charged bodies. It investigates net field strength, defines a function Y resembling a nuclear mass function, and suggests a nature's preference for heavier objects at the positive charge centre. It also finds an expression for ellipse geometry and introduces 'quintessence' analogous to electric potential. Finally, it explores the constancy of a factor χ (chi).

[2]
Date of completion: 20 Feb 2022
Date of pre-print/publication: 02 Oct 2023: OSF Preprints
Title: `The Coronal Heating Problem`
Type: Review Article
DOI: **10.31219/osf.io/63cag**
Abstract: An overview of the Sun's atmospheric temperature inversion and detailing the recent developments in resolving the coronal heating problem.

[1]
Date of completion: 08 Dec 2021
Date of pre-print/publication: 29 Sep 2023: OSF Preprints
Title: `Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle`
Guide: Dr. Himanshu Pandey, SVNIT-Surat -
Type: UG Project I, [Jul-Dec, 2021]
DOI: **10.31219/osf.io/ukx9t**
Abstract: This proposed theory explores two bodies' motion based on minimizing energy. By analyzing this ideal process, it suggests the existence of electric charges and gravity's role. It also finds a mass ratio for the bodies where the process works and supports the law of inertia.

Technical Projects

06/21-07/21	Internship, *The front-end development of a dummy e-commerce website* - Learning included using HTML5, CSS3 frameworks and training for web designing.
07/21-12/21	Project, *Building the student information management system using C language* -Learning included using C languages and its application in file handling

Courses

Ph.D. *self-preparation* /Undone:	Relativistic Quantum Mechanics, Classical Field Theory, Canonical Quantization, Interacting Quantum fields, Path Integrals*, General relativity I* & II, SUSY & Supergravity, Cosmology, String theory I & II, Axiomatic QFT, Basics of Topology and Differential Manifold, Differential Geometry, Algebraic Topology*, Lie Groups & Lie Algebra*, Groups & Representations*, Geometric Algebra*, Algebraic Geometry, Geometric group theory.
University (Physics):	Quantum Field Theory I, Many-Body Physics and Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Special Relativity, Quantum Mechanics I & II, Computational Physics(Python, Monte-Carlo), Computational Methods (MATLAB/Octave), Atomic and Molecular Physics, Electrodynamics, Statistical Mechanics, Astrophysics, Density Functional Theory, and Classical Mechanics.
University (Mathematics):	Singular perturbation theory, Special Functions, Linear Algebra, Tensors, Vector Calculus, Integral transforms, Real & Complex analysis, Computational Methods, Infinite Series, Solutions to ODEs and PDEs, Numerical methods, Interpolation, Partial Differentiation, solutions to linear equations systems etc.
Certifications:	A Course in Math History[Aug-Sep'21], Classical Electromagnetism[Aug-Dec'20], Learning Physics through Simple Experiments[Apr-Jun'20], The advanced course on Special Theory of Relativity[Jan-May'20], The basics of Quantum Mechanics[Aug-Nov'19] and The basics of Special Theory of Relativity[Jan'18-Mar'19], TCSiON Career Edge - TCS, Mumbai [6-16th Apr'20], Life Skills for Engineers (L1), CEMCA & University of Hyderabad [Jan-Feb'20],
Extra Courses:	Algebraic Topology (Prof. Prateep Chakraborty, Dec-23-*Present*), QFT-II (Prof. Partha Mukhopadhyay, Oct-23-*Present*), Group Theory Methods in Physics (Prof. P. Ramadevi, through Youtube.com, Aug-Sep'23), Introductory Lectures on Topology and Differential Geometry for Physicists (Prof. Sunil Mukhi, through Youtube.com, Sep-Oct'23)

Achievements

09/16	NTS Scholarship of merit, Government of India, New Delhi.
04/23	Qualified Graduate Aptitude Test in Engineering (GATE)-23, National Rank: 3409
02/20	Winner (Physics), InQuest 4.0, SCOSH [Student Chapter], SVNIT
03/20	Runner Up (Pratyaksha), Physics Club, SVNIT

Skills

Programme:	C, Python, MATLAB, Octave
Softwares:	LaTeX, MS Excel, Mathematica, and use of GitHub.
Web Design:	Basics of HTML5, CSS3
Languages:	English (Intermediate), Hindi(Native)

Workshops & Webinars

07/21	*Quantum Fields, Geometry & Representation Theory*, ICTS-TIFR, Bangalore.
12/20	National workshop on *Data analysis using MS Excel*, BBD NITM, Lucknow.
08/20	National workshop on MATLAB *Tools And Applications*, BBD NITM, Lucknow.

Talks & Books

02/07/23	`MaNoVighn`: Short Poetry Collections, Kindle, ASIN: B0C9Y7X1PG
01/09/21	`KālSanGharsha`: Short Poetry Collections, Kindle, ASIN: B09FC1XRPK
13/11/21	*Unification in Mathematics*, IntERAct Seminar series, AMHD, SVNIT-Surat

Responsibilities & Positions

08/22-03/23	**Student Coordinator**: QUANTA Seminar Series, Department of Physics, SVNIT Surat
11/22-01/23	**Design-Infra Coordination Head**: *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
10/21-07/22	**Joint Academic Affairs Secretary**: *Academic Affairs Council* SVNIT, Surat
03/21-07/22	**Co-convener/Co-head**: *Society for Cultivation of Science and Humanities* (SCOSH), Student Chapter, SVNIT Surat.
06/20-03/21	**Junior Graphic designer & Member**: *Society for Cultivation of Science and Humanities* (SCOSH) Student Chapter, SVNIT Surat
12/19-01/20	**Student Volunteer**: *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
09/19-07/20	**Student Volunteer**: *Unnat Bharat Abhiyan* (UBA) SVNIT Surat

Reference

Prof. Dr. Ajay Kumar Rai — email Id : `akr@phy.svnit.ac.in`— Professor, Physics, SVNIT

Dr. Vikash Kumar Ojha — email Id : `vko@phy.svnit.ac.in`— Assistant Professor, Physics, SVNIT

Declaration: *I hereby declare that the information given above is true to the best of my knowledge.*

Digitally Signed: `Roshan Raj`

Date: 04/02/2024.